

# Predictive Maintenance

End-to-end ML lifecycle on Azure ML + MLflow

**Goal: predict machine failure and manage the model from experiment tracking to deployment and monitoring.**



**3.4%**

failure rate

**0.974**

best CV ROC-AUC

**v1**

registered model

**Succeeded**

endpoint state

# Problem & Dataset

AI4I 2020 predictive-maintenance data: useful, imbalanced, and leakage-sensitive

## Why this is an MLOps problem

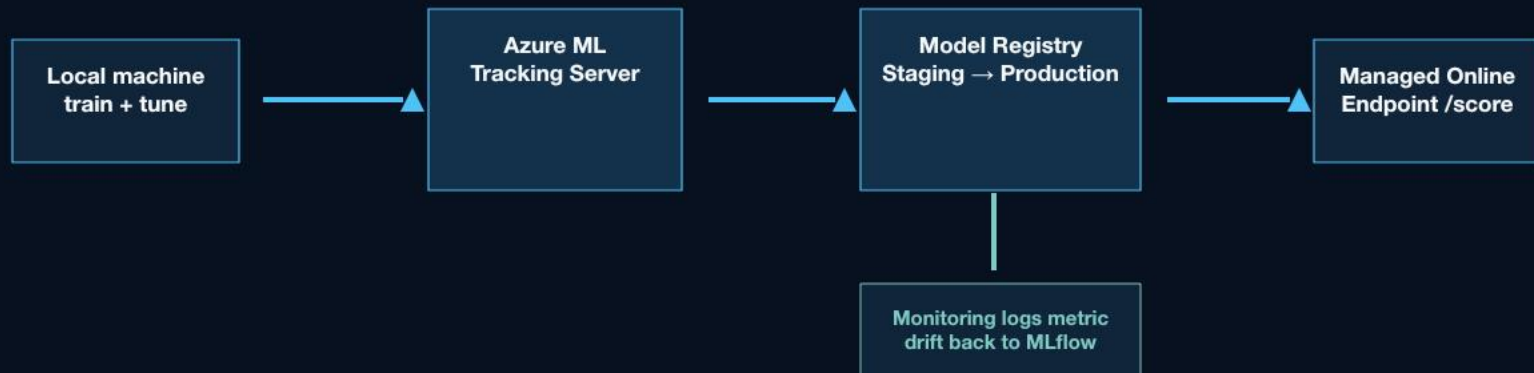
- Industrial failures create downtime; predictions must be reliable and repeatable.
- Only 3.4% of records are failures, so accuracy alone is misleading.
- Per-mode failure flags are dropped to avoid label leakage.
- Metrics focus on ROC-AUC, PR-AUC, recall, and F1.

Course link: ML as a process starts with the right problem framing, dataset, and success criteria.



# MLflow + Azure Architecture

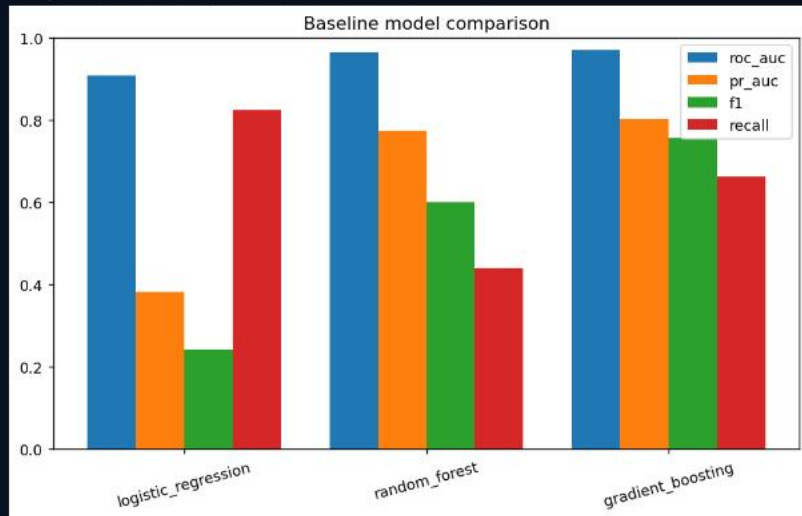
Local compute for cost control; Azure ML for shared lifecycle state



Azure credit was used where the course expects cloud MLOps: tracking, registry, and deployment.

# Experiments & Tuning

Every baseline and Optuna trial is traceable in MLflow



**20**

Optuna trials

**0.974**

Best CV ROC-AUC

**0.44 → 0.77**

Recall lift

## Logged per run

- parameters and hyperparameters
- ROC-AUC, PR-AUC, precision, recall, F1
- model artifact, signature, confusion matrix

# Registry & Deployment Demo

Best model registered, promoted, and deployed through Azure ML

**PredictiveMaintenanceModel**

Registered model

**1**

Version

**Staging → Production**

Lifecycle

**Succeeded**

Endpoint

Azure ML experiments / runs

Best run metrics + artifacts

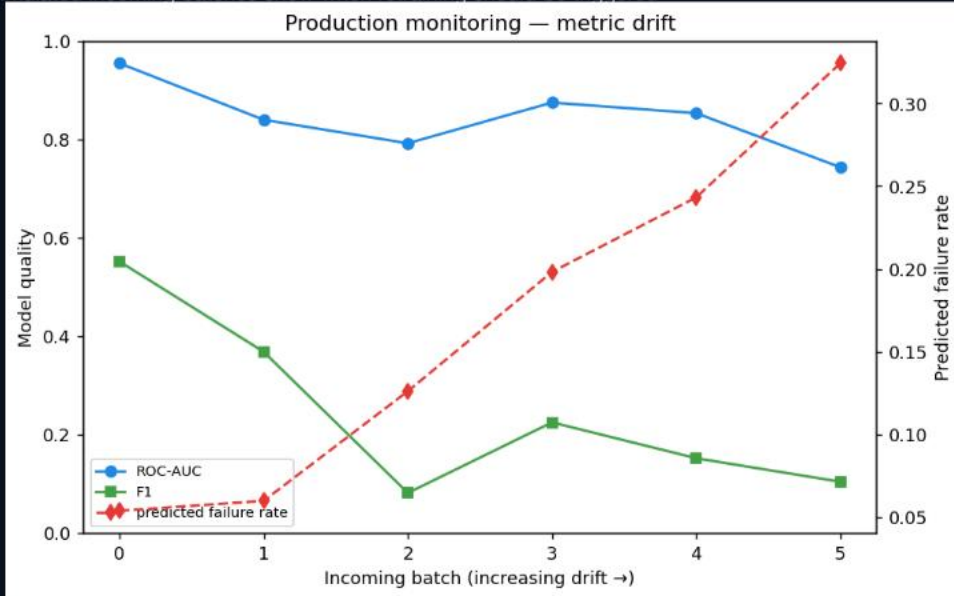
Model registry version 1

Endpoint status

Evidence captured: workspace mlw-predmaint · resource group rg-mlops-predmaint · endpoint predmaint-endpoint. Do not show scoring keys.

# Monitoring & Drift

Simulated incoming batches show when retraining should be triggered



**0.914**

Batch 0 ROC-AUC

**0.766**

Batch 5 ROC-AUC

**4% → 37%**

Predicted failure rate

Monitoring closes the lifecycle: drift is logged back to MLflow with step-indexed metrics.

# Conclusion: Full ML Lifecycle Delivered

The project satisfies the course objectives and deliverables

- ✓ **Experiment tracking: params, metrics, artifacts**
- ✓ **Model training + Optuna tuning**
- ✓ **MLflow model packaging + Azure endpoint**
- ✓ **Monitoring: drift simulation logged to MLflow**
- ✓ **Model registry: versioned Staging → Production**

## Deliverables

GitHub repo: public  
Project report: complete  
Presentation: PPTX + PDF backup  
Submission ZIP:  
PRJ-AyhanGurbangeldiyev-2020053.zip

**Main takeaway: this is not only a model score; it is a managed lifecycle.**